

1) Grey Level Transformation:

(a) Purpose of Grey Level Transformation.

→ Grey level transformation is used to improve the appearance of an image by changing pixel intensity values. It helps to enhance contrast, highlight details, or create special effects such as image negatives.

(b) Negative Transformation.

Formula:

$$s = 255 - r.$$

Wow?

Original Pixel (r)	Transformed Pixel (s)
50	205
100	155
150	105
200	55

Interpretation:

- Dark pixels become bright.
- Bright pixels become dark.
- The image appears as a photographic negative.

2) Histogram Equalization

Intensity	Frequency
0	4
1	6
2	10
3	6

$$\begin{aligned}\text{Total Pixels} &= 4 + 6 + 10 + 6 \\ &= 26.\end{aligned}$$

(a) Why Histogram Equalization is used.

→ Histogram Equalization improves image contrast by spreading pixel intensity values over the full intensity range.

(b) Cumulative Distribution Function (CDF):

Intensity	Frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

PDF?
formula?

(c) Equalized Intensity values:

Formula:

$$s = \frac{\text{CDF}}{26} \times (4-1)$$

Intensity	calculation	New value.
0	$\frac{4}{26} \times 3 = 0.46$	0
1	$\frac{10}{26} \times 3 = 1.15$	1
2	$\frac{20}{26} \times 3 = 2.31$	2
3	$\frac{26}{26} \times 3 = 3$	3

Equalized values : 0, 1, 2, 3.

3) Gaussian Smoothing:

Gaussian Kernel:

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image Region:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

(a) Why Gaussian Filtering is Preferred.

- Reduces noise smoothly.
- Preserves edges better than averaging filters.
- Gives more weight to the centre pixel.

(b) New Centre Pixel.

$$\begin{aligned} &= \frac{1}{16} (10 + 40 + 10 + 40 + 160 + 40 + 10 + 40 + 10) \\ &= \frac{360}{16} \\ &\Rightarrow 22.5 \end{aligned}$$

$$\text{New Center Pixel} = 22.5 \\ \Rightarrow (\approx 23).$$

4) Edge Detection using Sobel operator.

Horizontal Sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

(a) Purpose of Edge Detection

→ Edge detection identifies object boundaries by detecting sudden intensity changes.

(b) Gradient calculation.

$$= (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + (0) + (1 \times 30) + (2 \times 30) + (1 \times 30).$$

$$= -10 - 20 - 10 + 30 + 60 + 30$$

$$\Rightarrow 80.$$

→ Gradient at center = 80.

5) Contrast stretching.

(a) Need for contrast stretching.

→ Contrast stretching increases the differences between dark and bright pixels, making image details more visible.

(b) Linear contrast stretching.

Formula:

$$s = \frac{r - 50}{150 - 50} \times 255$$

For $r = 50$.

$$s = \frac{50 - 50}{100} \times 255 = 0.$$

For $r = 100$

$$s = \frac{100 - 50}{100} \times 255 = 127.5$$

$$s \approx 128.$$

For $r = 150$

$$s = \frac{150 - 50}{100} \times 255$$

$$s = 255$$

original value	new value
50	0
100	128
150	255